

Hoang Anh Vu

Boston, MA | 737-326-8348 | anhvu@bu.edu | linkedin.com/in/anhvh1299/ | github.com/hoangvu05

Education

Boston University

B.A in Computer Science, B.A in Psychology

- **GPA: 3.8/4.0** | Dean's List (all semesters)
- **Coursework:** Data Structures and Algorithm, Functional Programming, Computer Vision, Computer Systems, Probability, Discrete Math

Expected May 2027

Boston, Massachusetts

Technical Skills

- **Languages:** Python, Java, C, C++, C#, JavaScript, OCaml, R, HTML, CSS
- **Developer Tools:** SQL Server, MySQL, NoSQL, Google Cloud Platform, Vercel, AWS (S3), WSL, React, Express.js
- **Technologies/Frameworks:** Firebase, Git, GitHub, Spring Boot, .NET, Figma

Experience

Boston University, Department of Biology

Quantitative Biology Research Assistant

Mar 2025 – Present

Boston, MA

- Perform research on molecular mechanisms and regulation of protein synthesis in human cells by testing software packages to understand the leading cause of neurodegenerative disorders.
- Implement alignment workflows using Bowtie2 and SAMtools on a WSL2 environment, use Python to map and visualize over 60 million short gene reads to specific RNA transcript to understand the translation initiation regulation.
- Utilize R for data extraction, quality control, and interpret complex sequencing data and assess hypotheses regarding translational efficiency.

Spark! Innovation Fellowship

Innovation Fellow

Jan 2025 – May 2025

Boston, MA

- Developed MarkItIt, a platform to streamline the process of organizing pop-up events for small businesses and vendors with 300+ vendors using currently, under the guidance of industry mentors through the fellowship and got 2nd place.
- Built the backend in Firebase, frontend in React, and deployment in Vercel.
- Implemented end-to-end features on weekly Agile Scrum task using analyzed insights from 100+ interviews and surveys to identify a flawed customer journey to improve the application. Shipped 12 full-stack iterations total.
- Used Puppeteer to web-scrape 1000s pop-up events and built an AI algorithm for event matching and calculate finances.
- Collaborated with a team of 5 using Agile methodologies and MoSCoW method, employed Slack, Notion, Git, and Github for project management and version control.

Crown X, a division of Masan Group

Software Engineer Intern

Jun 2024 – Aug 2024

Ho Chi Minh City, Vietnam

- Engineered a full-stack ASP.NET Core MVC web application to modernize a legacy supply chain system. The solution simplify bottleneck pipeline, reduced operational latency by 25% and prevented errors in inventory tracking.
- Developed the application using C#, JavaScript, and SQL Server, creating a centralized catalog for a database of 1,000+ e-commerce items, enhancing data accessibility for internal teams.

Projects

Companion

Dec 2024 – Present

- Developed a cloud web-based senior care management app with Java, Spring Boot, and MySQL for the back-end, and HTML/CSS/JavaScript front-end. Tested RESTful APIs with Postman.
- Integrate features such as member administration, push notifications, caregiver reminders, text-to-speech with Google API, and story creation with OpenAI API.
- Deploy the app on Google Cloud Platform for scalability and QA tested the app on 10+ users.
- Collaborate with a team of 5 using Agile methodologies, employing Teams, Jira, and Git for project management and version control. Do weekly SCRUM task and biweekly internal meeting.

College Carpool

Nov 2024 – Present

- Win \$500 in Boston University Sustainability Seed Grant to develop a matchmaking system to recommend student passengers possible student drivers, factoring fuel costs, distance of travel, etc., for a carpooling service for students.
- Join the development with 3+ engineers using Firebase, Websocket, Express.js, NoSQL in the backend for real time communication with minimal latency, database and transaction for ACID completeness and concurrency control.
- Use React for the frontend and deploy in Vercel, while working closely with BU IT to integrate the university's OAuth.

Extracurricular & Achievements

- *Boston University HackHardware Hackathon*, 2nd place for making a controller using Arduino Nov 2024
- *Boston University Model United Nation*, Assistant Crisis Director, BosMUN XXIII (Congress of Vienna) Feb 2024
- *Ho Chi Minh City Team Programming Competition for Youths*, 1st place team Mar 2023